RISMS – Code & Branch Coverage Report

1. Method

Coverage was measured with gcov (g++ `--coverage`, `-O0`) and reported with gcovr 8, using `--exclude-throw-branches` and `--exclude-unreachable-branches` so compiler-generated exception edges are not counted. Two instrumented builds were run: the unit test binary (logic layer only) and the full `risms` binary driven by `tests/system_tests.sh`. Their results were then merged.

2. Overall Coverage

Test level	Scope	Lines	Functions	Branches
Unit tests	Logic layer (11 files, excl. `ConsoleUI`, `MenuController`, `main`)	91.4% (604/661)	94.9% (74/78)	83.3% (547/657)
System tests	All 14 files	85.2% (922/1,082)	92.0% (103/112)	77.2% (940/1,217)
Combined	**All 14 files**	**90.7% (981/1,082)**	**94.6% (106/112)**	**83.0% (1,010/1,217)**

3. Coverage by File (combined)

File	Lines	Line %	Branches	Branch %
ReportService.cpp	27/27	100%	12/12	100%
AuthService.cpp	58/58	100%	54/61	88.5%
ConfigStore.cpp	42/42	100%	36/45	80.0%
Validator.cpp	25/26	96.2%	34/37	91.9%
ProductService.cpp	28/29	96.6%	20/22	90.9%
MenuController.cpp	283/299	94.6%	349/396	88.1%
LedgerRepository.cpp	90/99	90.9%	85/97	87.6%
ProductRepository.cpp	113/125	90.4%	107/121	88.4%

File	Lines	Line %	Branches	Branch %
StockService.cpp	84/93	90.3%	64/80	80.0%
UserRepository.cpp	61/68	89.7%	58/67	86.6%
FileStore.cpp	54/65	83.1%	60/92	65.2%
main.cpp	39/47	83.0%	52/67	77.6%
errorlogger.cpp	22/29	75.9%	18/23	78.3%
ConsoleUI.cpp	55/75	73.3%	61/97	62.9%
Total	**981/1,082**	**90.7%**	**1,010/1,217**	**83.0%**

4. Uncovered Code

File (lines)	What is not covered	Why	Action
FileStore.cpp (21, 52, 68, 75–91, 109)	Directory-create, temp-file open, flush and rename failures; append open failure	Needs OS-level faults (full disk, permissions)	Accept; partly exercised by UT-43
StockService.cpp (59, 86, 123, 147, 169–180)	Product save failure; stock-out ledger failure; rollback-also-fails path	Needs a failing write mid-operation	Add a stock-out version of UT-43
main.cpp (34, 69–94)	Seed-user failure; exit on unreadable users, products or ledger file	Requires corrupt or unreadable startup files	Accept
ConsoleUI.cpp (29–30, 41–42, 67–82, 108–122)	`readDouble()` (never called); recoverable stream error; whitespace-only input; Windows/terminal password branches	Dead code and platform-specific code	Remove `readDouble()`; add a whitespace-only input system test
MenuController.cpp (296–305)	`handleFindById()`	Not reachable from any menu option (see DEV-02)	Wire to a “Search by ID” option or delete

File (lines)	What is not covered	Why	Action
MenuController.cpp (179–180, 200–201)	Blank price / threshold in Add Product	Not scripted	Add a system test pressing Enter at those prompts
MenuController.cpp (486–487)	Error branch of Inventory Summary	`inventorySummary()` never fails	Accept
UserRepository.cpp (28, 38–41, 52, 58, 92)	Malformed user record on load; user save failure	Not tested	Add a unit test loading a bad `users.txt`
errorlogger.cpp (40–45, 54–59, 70)	`info()`, `error()` and log-file open failure	Only `warning()` is triggered by tests	Accept